

Configuración https

1. Actualizar la instancia

```
sudo apt update && sudo apt upgrade -y
```

```
root@ip-172-31-70-217:/home/ubuntu# apt update && apt upgrade -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1675 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [309 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Component s [175 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Met
```

2. Instalar Apache y módulos SSL

```
sudo apt install apache2 openssl -y
sudo a2enmod ssl
sudo systemctl restart apache2
```

```
root@ip-172-31-70-217:/home/ubuntu# root@ip-172-31-70-217:/home/ubuntu# apt install ap
che2 openssl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.8).
openssl is already the newest version (3.0.13-0ubuntu3.6).
The following packages were automatically installed and are no longer required:
  linux-aws-6.14-headers-6.14.0-1015 linux-aws-6.14-tools-6.14.0-1015
  linux-headers-6.14.0-1015-aws linux-image-6.14.0-1015-aws
  linux-modules-6.14.0-1015-aws linux-tools-6.14.0-1015-aws
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@ip-172-31-70-217:/home/ubuntu# a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Module socache_shmcb already enabled
Module ssl already enabled
```

3. Crear directorio para los certificados

```
mkdir -p /etc/ssl/dani
```

```
root@ip-172-31-70-217:/home/ubuntu# mkdir -p /etc/ssl/dani
```

4. Generar certificado autofirmado

```
openssl req -x509 -nodes -days 365 \ -newkey rsa:2048 \ -keyout
/etc/ssl/gema/server.key \ -out /etc/ssl/gema/server.crt
```

```
root@ip-172-31-70-217:/home/ubuntu# openssl req -x509 -nodes -days 365 \
> -newkey rsa:2048 \
> -keyout /etc/ssl/gema/server.key \
> -out /etc/ssl/gema/server.crt
.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+
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-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:
State or Province Name (full name) [Some-State]:
Locality Name (eg, city) []:
Organization Name (eg, company) [Internet Widgits Pty Ltd]:
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:asir.com
Email Address []:
root@ip-172-31-70-217:/home/ubuntu#
```

5. Crear archivo de VirtualHost para HTTPS

```
sudo nano /etc/apache2/sites-available/asir-ssl.conf
```

```
root@ip-172-31-70-217:/home/ubuntu
GNU nano 7.2 /etc/apache2/sites-available/asir-ssl.conf
<VirtualHost *:443>
  ServerName asir.com
  DocumentRoot /var/www/html
  SSLEngine on
  SSLCertificateFile /etc/ssl/gema/server.crt
  SSLCertificateKeyFile /etc/ssl/gema/server.key
  <Directory /var/www/html>
    Options Indexes FollowSymLinks
    AllowOverride All
    Require all granted
  </Directory>
</VirtualHost>
```

6. Habilitar sitio SSL y deshabilitar el default

```
sudo a2dissite default-ssl.conf
sudo a2ensite asir-ssl.conf
```

```
sudo systemctl reload apache2
```

```
root@ip-172-31-70-217:/home/ubuntu# a2dissite default-ssl.conf
Site default-ssl already disabled
root@ip-172-31-70-217:/home/ubuntu# a2ensite asir-ssl.conf
Site asir-ssl already enabled
root@ip-172-31-70-217:/home/ubuntu# systemctl reload apache2
root@ip-172-31-70-217:/home/ubuntu#
```

comprobar apache que reconoce el Virtualhost:

```
sudo apache2ctl -S
```

```
root@ip-172-31-70-217:/home/ubuntu# sudo apache2ctl -S
VirtualHost configuration:
*:80 ip-172-31-70-217.ec2.internal (/etc/apache2/sites-enabled/000-default.conf:1)
*:443 is a NameVirtualHost
       default server alumno1.590183782760-claseawsacademy.com (/etc/apache2/sites-enabled/alumno1-ssl.conf:1)
       port 443 namevhost alumno1.590183782760-claseawsacademy.com (/etc/apache2/sites-enabled/alumno1-ssl.conf:1)
       port 443 namevhost asir.com (/etc/apache2/sites-enabled/asir-ssl.conf:1)
ServerRoot: "/etc/apache2"
Main DocumentRoot: "/var/www/html"
Main ErrorLog: "/var/log/apache2/error.log"
Mutex default: dir="/var/run/apache2/" mechanism=default
Mutex watchdog-callback: using_defaults
Mutex ssl-stapling-refresh: using_defaults
Mutex ssl-stapling: using_defaults
Mutex ssl-cache: using_defaults
PidFile: "/var/run/apache2/apache2.pid"
Define: DUMP_VHOSTS
Define: DUMP_RUN_CFG
User: name="www-data" id=33
Group: name="www-data" id=33
root@ip-172-31-70-217:/home/ubuntu#
```

7. Configurar resolución de dominio dentro de la VPC

En cada máquina cliente (o la misma EC2):

```
sudo nano /etc/hosts
172.31.68.185 asir.com
```

Agregar línea (ip privada de tu EC2):

```
172.31.70.217 asir.com
```

```
GNU nano 7.2 /etc/hosts *
127.0.0.1 localhost
172.31.70.217 asir.com
# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

8. Probar HTTPS:

```
curl -vk https://asir.com
```

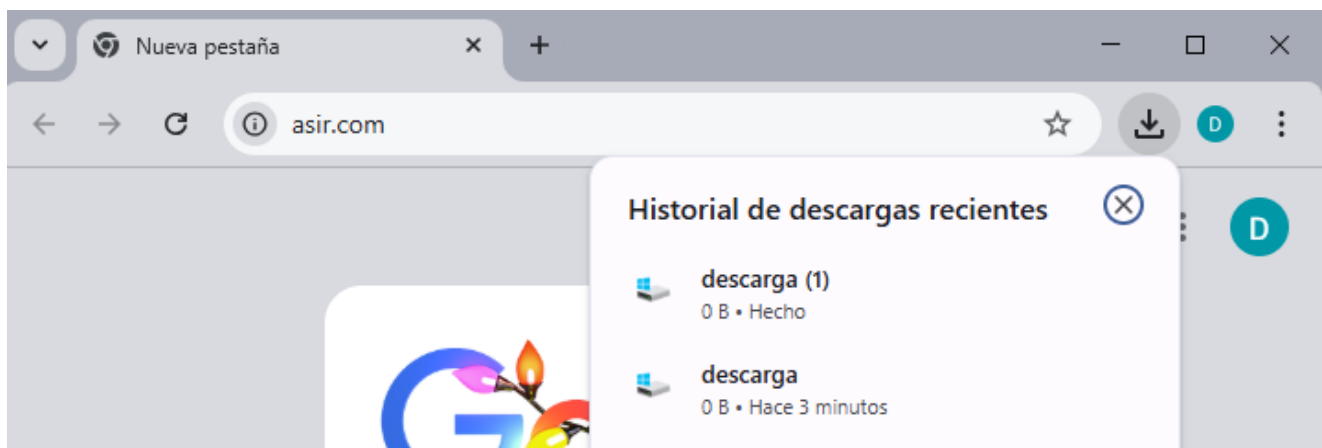
```

root@ip-172-31-70-217:/home/ubuntu# curl -vk https://asir.com
* Host asir.com:443 was resolved.
* IPv6: (none)
* IPv4: 172.31.70.217
*   Trying 172.31.70.217:443...
* Connected to asir.com (172.31.70.217) port 443
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_256_GCM_SHA384 / X25519 / RSASSA-PSS
* ALPN: server accepted http/1.1
* Server certificate:
*  subject: C=AU; ST=Some-State; O=Internet Widgits Pty Ltd; CN=asir.com
*  start date: Dec 12 17:58:40 2025 GMT
*  expire date: Dec 12 17:58:40 2026 GMT
*  issuer: C=AU; ST=Some-State; O=Internet Widgits Pty Ltd; CN=asir.com
*  SSL certificate verify result: self-signed certificate (18), continuing anyway.
*  Certificate level 0: Public key type RSA (2048/112 Bits/secBits), signed using sha2
56WithRSAEncryption
* using HTTP/1.x
> GET / HTTP/1.1
> Host: asir.com
> User-Agent: curl/8.5.0
> Accept: */*
>
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
* old SSL session ID is stale, removing
< HTTP/1.1 200 OK
< Date: Fri, 12 Dec 2025 18:09:17 GMT
< Server: Apache/2.4.58 (Ubuntu)
< Last-Modified: Fri, 05 Dec 2025 18:33:15 GMT
< ETag: "51f-64538adb7522e"
< Accept-Ranges: bytes
< Content-Length: 1311
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Servidor Apache</title>
  <style>

```

▼ Hoy (3)

 descarga	12/12/2025 19:10	Archivo	0 KB
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9. Redirigir HTTP → HTTPS

Editar archivo http:

```
sudo nano /etc/apache2/sites-available/000-default.conf
```

Agregar dentro:

```
Redirect "/" "https://asir.com/"
```

```
<VirtualHost *:80>
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html
    Redirect "/" "https://asir.com/"

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

```
sudo systemctl reload apache2
```

10. Evidencia:

```
curl -I https://asir.com -k
curl -v -k http://asir.com
```

```
root@ip-172-31-70-217:/home/ubuntu# curl -I https://asir.com -k
HTTP/1.1 200 OK
Date: Fri, 12 Dec 2025 18:19:59 GMT
Server: Apache/2.4.58 (Ubuntu)
Last-Modified: Fri, 05 Dec 2025 18:33:15 GMT
ETag: "51f-64538adb7522e"
Accept-Ranges: bytes
Content-Length: 1311
Vary: Accept-Encoding
Content-Type: text/html
```

```
root@ip-172-31-70-217:/home/ubuntu# curl -v -k http://asir.com
* Host asir.com:80 was resolved.
* IPv6: (none)
* IPv4: 172.31.70.217
* Trying 172.31.70.217:80...
* Connected to asir.com (172.31.70.217) port 80
> GET / HTTP/1.1
> Host: asir.com
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 302 Found
< Date: Fri, 12 Dec 2025 18:20:06 GMT
< Server: Apache/2.4.58 (Ubuntu)
< Location: https://asir.com/
< Content-Length: 275
< Content-Type: text/html; charset=iso-8859-1
<
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>302 Found</title>
</head><body>
<h1>Found</h1>
<p>The document has moved <a href="https://asir.com/">here</a>.</p>
<hr>
<address>Apache/2.4.58 (Ubuntu) Server at asir.com Port 80</address>
</body></html>
* Connection #0 to host asir.com left intact
root@ip-172-31-70-217:/home/ubuntu#
```

11. Descargar el certificado a Windows

Localiza el certificado en la EC2:

```
root@ip-172-31-70-217:/home/ubuntu# ls /etc/ssl/gema/server.crt
/etc/ssl/gema/server.crt
```

Copiar el certificado a mi máquina Windows:

```
scp -i "server-route.pem" ubuntu@:/etc/ssl/gema/server.crt C:\Users\\Downloads\
```



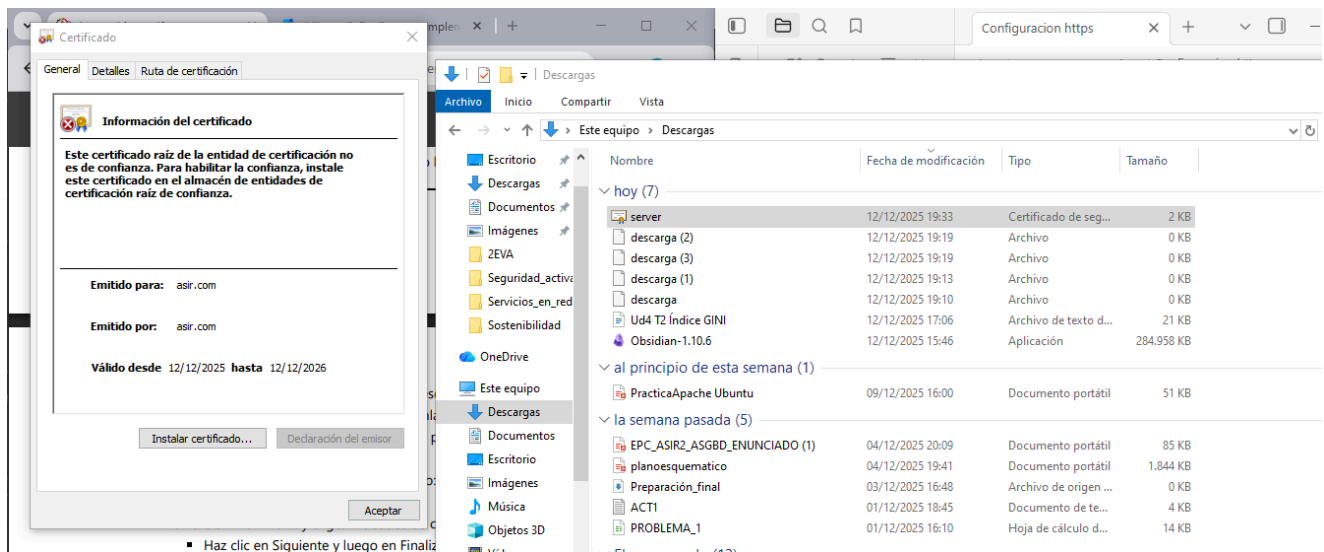
```
C:\Users\2ASIR\Documents\Backup_clases_2ASIR\Arquitectura_nube\claves-aws>scp -i "server-route.pem" ubuntu@34.232.179.36:/etc/ssl/gema/server.crt C:\Users\2ASIR\Downloads
server.crt
100% 1298 6.0KB/s 00:00
```

Comprobar que se ha pasado el certificado correctamente:

```
root@ip-172-31-70-217:/etc/ssl/gema# ls -la
total 16
drwxr-xr-x 2 root root 4096 Dec 12 17:58 .
drwxr-xr-x 7 root root 4096 Dec 12 17:55 ..
-rw-r--r-- 1 root root 1298 Dec 12 17:58 server.crt
-rw----- 1 root root 1704 Dec 12 17:57 server.key
root@ip-172-31-70-217:/etc/ssl/gema#
```

12. Instalar el certificado en Windows

Abrir el certificado:



Instalar el certificado:

